Assignment No 4

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Title – Implementation of Tower of Hanoi Using State Space Search.

Program:-

```
def tower_of_hanoi(n, source, auxiliary, destination):
    if n == 0:
        return

    # Move n-1 disks from source to auxiliary
    tower_of_hanoi(n - 1, source, destination, auxiliary)

    # Move the largest disk to destination
    print(f"Move disk {n} from {source} -> {destination}")

    # Move n-1 disks from auxiliary to destination
    tower_of_hanoi(n - 1, auxiliary, source, destination)


# Main program
n = int(input("Enter number of disks: "))
```

```
print("\nTower of Hanoi using State Space Search:")

tower_of_hanoi(n, 'A', 'B', 'C')


print("\nMinimum number of moves:", 2**n - 1)
```

Output:-

```
Enter number of disks: 3


Tower of Hanoi using State Space Search:

Move disk 1 from A -> C

Move disk 2 from A -> B

Move disk 1 from C -> B

Move disk 3 from A -> C

Move disk 1 from B -> A

Move disk 2 from B -> C

Move disk 1 from A -> C


Minimum number of moves: 7


Process finished with exit code 0
```